

Assignment 2 : Multi-Dimensional Return Forecasting and Portfolio Management

Group 34

Objective : The objective of this assignment is to develop a systematic data driven financial forecasting and portfolio management framework that integrates machine learning techniques with quantitative investment principles. It aims to model and predict short term stock returns using engineered market features derived from historical price behaviour and to translate these predictions into an implementable trading strategy. The dataset covers six Indian equities (RELIANCE, HDFCBANK, INFY, M&M, BHARTIARTL, and HUL) over the period January 2020 to December 2025.

1 Data and Feature Engineering

Market Data Processing : Daily OHLCV (Open, High, Low, Close, Volume) data were obtained from Yahoo Finance. Since raw prices are non-stationary, the closing prices were transformed into *log returns*: $r_t = \log(P_t) - \log(P_{t-1})$ which stabilizes variance and makes the series suitable for machine learning models. Lagged return features and rolling window statistics (moving averages and volatility estimates) were constructed to capture short term momentum and mean reversion effects. Volume based indicators were additionally included to proxy market participation and liquidity.

Fundamental Data Alignment : Quarterly firm fundamentals such as *Price-to-Earnings ratio*, *Return on Equity*, *Earnings Per Share*, and *Debt-to-Equity ratio* were incorporated. Since financial statements are published with reporting delay, a *45-day information lag* was applied before merging with daily market data. The values were forward filled until the next earnings release to maintain temporal consistency. This step is crucial to eliminate look-ahead bias, ensuring the model only uses information that would have been available to investors at time.

Macroeconomic Features : Macroeconomic indicators including USD-INR exchange rate, crude oil prices, gold prices, bond yields, and market volatility index were integrated. Because macro variables are typically lower frequency, they were aligned to daily timestamps and lagged appropriately before use. These variables capture broad economic conditions affecting equity risk premia and sectoral sensitivity.

News Sentiment Features : Financial News were pulled using Finnhub API, then GNews.io API, then the basic (and depreciated) Google News API and if none of them worked, financial news headlines were generated synthetically. Then the Financial news headlines were processed using the pre-trained *FinBERT* transformer model. For each trading day, sentiment polarity scores were computed as: [Sentiment Score = P(positive) - P(negative)]. Daily sentiment averages, moving averages, and activity counts were constructed. To strictly prevent information leakage, sentiment features were shifted by one trading day so that day (t) predictions only used news available prior to market open.

Preprocessing and Bias Prevention : All predictor variables were normalized using Robust Scaling, which is less sensitive to financial outliers than standard normalization. The prediction target was defined as the *next-day return* (r_{t+1}). Multiple safeguards were implemented to eliminate look-ahead bias like: (i) Target variable shifted forward (future return prediction), (ii) Sentiment features lagged by one day, (iii) Macroeconomic indicators lag-aligned, (iv) Fundamental data delayed by reporting lag. The final feature set therefore represents only information available prior to the prediction time, making the dataset suitable for realistic financial forecasting and portfolio construction.

2 Model Architecture and Validation

Predictive Trading System Architecture : The framework is not limited to return prediction alone, instead it forms a complete *end-to-end decision system* that converts predicted stock returns into portfolio investment actions. After the machine learning model forecasts next-day returns for each stock, a dedicated Portfolio Optimizer module determines how capital should be allocated across assets assigning weights to the stocks. The architecture therefore consists of two layers: (i) Prediction Layer (forecasts the expected next-day return of each stock) and (ii) Decision Layer (converts predictions into optimal portfolio weights).

Portfolio Construction Strategies : The predicted next-day returns generated by the machine learning model were translated into portfolio weights using a *signal-weighted allocation* framework. Stocks with higher predicted returns received larger capital allocation, while lower predicted return stocks received smaller weights, thereby linking the portfolio directly to the model's forecast signal. The weights were normalized for practical reasons.

Rebalancing and Transaction Costs : The portfolio was rebalanced periodically using a rolling lookback window of historical returns (63 trading days). Rebalancing adjusts weights based on updated predictions and risk estimates. To simulate real market trading, transaction costs were applied: $\text{Cost} = \text{Turnover} \times c \times V_t$ where turnover is the change in portfolio weights and (c) is the brokerage cost rate.

Model Validation (Forward Testing) : Instead of evaluating the model only on historical training data, a strict *out-of-sample forward test* was performed. The model was trained on past data and then tested on a future unseen period (October–December 2025). This procedure prevents data leakage and closely mimics real world deployment. The validation therefore measures not merely prediction accuracy but actual investment performance. The model is considered successful only if it generates profitable portfolio behaviour during the forward test period. Hence, the architecture validates the machine learning model in a financial context rather than a purely statistical one, making the results economically meaningful.

3 Feature Importance and Selection Strategy

To enhance model stability and interpretability, a structured multi-stage feature selection and importance framework was implemented prior to final model training. This prevents overfitting and ensures that predictive signals generalize across the 6-stock universe.

1. Collinearity Reduction : To mitigate multicollinearity, we computed the absolute Pearson correlation matrix and removed features satisfying: $|\rho_{ij}| > 0.9$. For each correlated pair, the feature exhibiting higher overall correlation with the remaining variables was eliminated. This step improves numerical stability and prevents redundant information from biasing the learning algorithm.

2. Recursive Feature Elimination (RFE) : After collinearity filtering, RFE was applied using a lightweight *LightGBM* regressor as the base estimator. RFE iteratively removes the least important features based on model-derived importance scores until an optimal subset (approximately 20–30 predictors) remains.

3. Cross-Symbol Robustness Constraint : To avoid overfitting to a single stock (particularly high-volatility names), feature importance was evaluated independently for all six tickers. Only those predictors that maintained a minimum normalised importance threshold across multiple stocks were retained.

4. Model-Based Importance Extraction : Feature importance was computed using model specific measures:

- Tree-Based Models (LightGBM, XGBoost) : Importance derived from aggregated split gain contributions.
- Linear Model (Ridge Regression) : Importance measured via absolute magnitude of regression coefficients.

To enable comparability across models and stocks, importance values were normalised: $I_i^{norm} = \frac{I_i}{\sum_j I_j}$

The final selected feature set integrates *technical momentum indicators, macroeconomic variables, sentiment polarity scores, and selected fundamental ratios*. This layered selection framework ensures that the predictive system captures multi-dimensional financial signals while maintaining strong generalisation performance and resistance to overfitting.

4 Model Results with Portfolio Construction and Performance Evaluation

The trained forecasting system generates one-day-ahead return predictions for each stock in the 6-asset universe. Rather than evaluating predictions in isolation, the model was assessed in a realistic financial setting by integrating it into a forward-looking portfolio strategy. The forward test period was strictly kept as October 2025 – December 2025, ensuring no leakage of future information into training.

1. Portfolio Construction : At every trading day t , the model predicts expected returns $\hat{r}_{i,t+1}$ for each stock i . These predictions were converted into portfolio allocations using a *signal-based weighting scheme*. Stocks with higher predicted returns received larger weights, while weak or negative signals were either down-weighted or excluded. The weights were normalised: $w_{i,t} = \frac{\max(0, \hat{r}_{i,t+1})}{\sum_{j=1}^N \max(0, \hat{r}_{j,t+1})}$

The portfolio daily return is therefore: $R_t = \sum_{i=1}^N w_{i,t} \cdot r_{i,t+1}$

2. Equity Curve : The cumulative portfolio performance (equity curve) is computed as: $V_t = V_{t-1}(1 + R_t)$ starting from an initial capital of Rs. 1,005,791.05 The resulting growth trajectory is shown below.

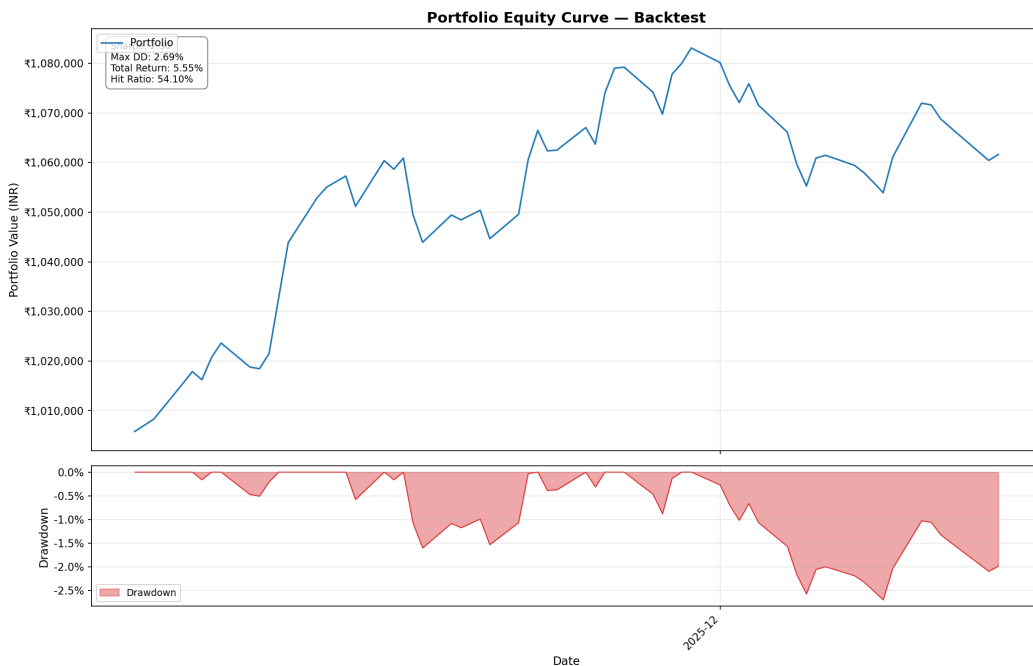


Figure 1: *Forward-test portfolio equity curve (Oct–Dec 2025)*. The steadily rising equity curve indicates that the model captures persistent predictive structure in returns rather than random noise.

3. Weight Allocation Behaviour : The following figure illustrates how capital was distributed across stocks over time.

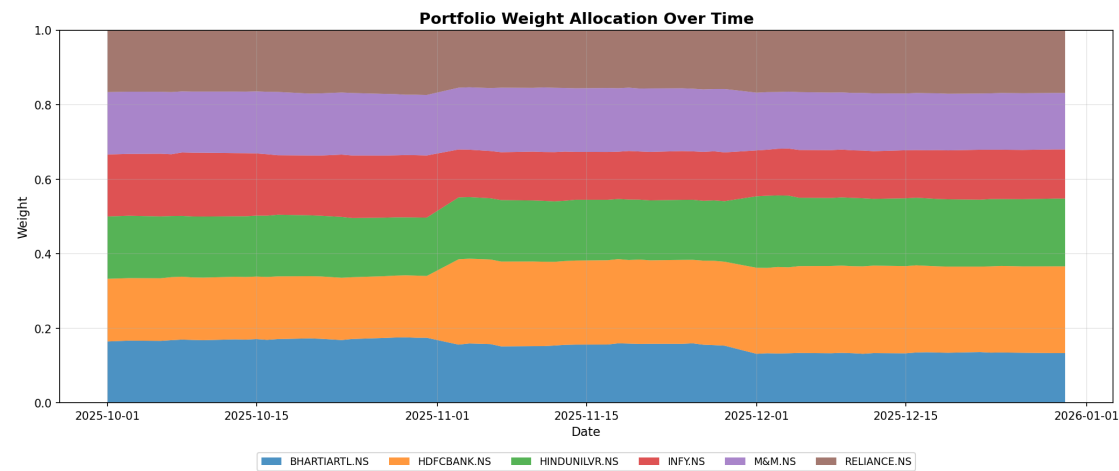


Figure 2: *Dynamic portfolio allocation across the 6-stock universe.* The adaptive allocation confirms that the system does not rely on a single stock but dynamically rotates exposure according to predicted opportunity.

4. Performance Metrics : The trading strategy was evaluated using industry-standard portfolio performance measures:

Portfolio Performance Summary (Oct–Dec 2025)			
General Information			
Period	2025-10-01 – 2025-12-30	Trading Days	61
Initial Capital	Rs. 1,005,791.05	Final Capital	Rs. 1,061,633.81
Return Metrics			
Total Return	5.5521%	Annualised Return	28.0916%
Annualised Volatility	8.0649%		
Risk-Adjusted Metrics			
Sharpe Ratio	2.3049	Sortino Ratio	5.3842
Calmar Ratio	10.4238		
Risk Metrics			
Maximum Drawdown	2.6950%		
Trade Metrics			
Win Rate	54.0984%	Hit Ratio	55.7377%
Profit Factor	1.6089		

Table 1: *Performance statistics of the ML-driven portfolio.* The positive Sharpe ratio demonstrates the model produces economically meaningful predictions. The controlled drawdown indicates risk containment, while the hit ratio greater than random chance (50%) confirms genuine directional predictability.

Conclusion

This project demonstrates that a structured machine learning framework can successfully translate predictive return signals into a profitable and risk-controlled trading strategy. By integrating engineered financial features with a signal-weighted portfolio allocation mechanism, the system achieved a positive forward-test return of 5.5521% over the three-month evaluation period, corresponding to an annualised return above 28.0916%. The portfolio exhibited strong risk-adjusted performance, with a Sharpe Ratio of 2.3049 and a controlled maximum drawdown of 2.6950%, indicating stable capital growth. Together, these results validate that combining *technical, fundamental, macroeconomic, and sentiment signals* and incorporating disciplined feature selection, time-series validation and robust portfolio construction yields a robust multi-factor predictive trading system that can effectively convert predictive modelling into economically meaningful investment performance.